



BUDGE BUDGE INSTITUTE OF TECHNOLOGY

Nischintapur, Budge Budge, Kolkata-700137

Department of Computer Science and Engineering/AI/IT

Academic Session: 2024-2025 (EVEN Semester)

Subject: Computer Networks Lab

Subject Code: PCC-CS692

SET-1

1. Change your host computer's ip to 172.45.2.45
2. You are unable to access the MAKUT website and want to quickly check whether the University's site "www.makautexam.net" is reachable from your system. what command will help you test the connection?
3. Write a TCP client-server C program where a Microsoft client sends "Hello Microsoft Support" and the server replies "Welcome to Microsoft Helpdesk" using port 9000, buffer size 512 bytes, and server IP as INADDR_ANY.

Question Number.	Course Outcome(CO)	Taxonomy levels
Question No-1	CO1	3
Question No - 2	CO1	3
Question No - 3	CO3	6



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SET2

1. Change your host computer's ip to 192.45.6.147
2. Your internet is slow, and you suspect delays are happening along the network path while connecting to amazon.com; which command can help you identify the intermediate routers and the delay at each hop?
3. Create a TCP client-server C program where an Amazon Alexa client sends "Get Weather Update" and the server responds "Today's forecast: Sunny and Clear" using port 7070, buffer size 2048 bytes, and server IP 127.0.0.1.

Question Number.	Course Outcome (CO)	Taxonomy levels
Question No-1	CO1	3
Question No - 2	CO1	4
Question No - 3	CO3	6



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SET 3

1. Change your host computer's ip to 192.168.2.65
2. A system in your college lab cannot connect to the internet; as a network administrator, which command would you use to verify the local machine's IP address, subnet mask, and gateway settings?
3. Develop a TCP client-server program in C where a Google Home device sends "Initiate Device Pairing" and the Google server replies "Pairing Successful" using port 6006, buffer size 1024 bytes, and server IP 192.168.2.65.

Question Number.	Course Outcome (CO)	Taxonomy levels
Question No-1	CO1	3
Question No - 2	CO1	3
Question No - 3	CO3	6



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SET4

1. Change your host computer's ip to 160.54.25.36
2. You suspect a malicious application is listening on a port in your Ubuntu system; which command would you use to list all active network connections and the ports currently in use?
3. Implement a TCP client-server C program where an Infosys employee client sends "Raise IT Ticket" and the server replies "Ticket Acknowledged" using port 5050, buffer size 1500 bytes, and server IP as INADDR_ANY.

Question Number.	Course Outcome (CO)	Taxonomy levels
Question No-1	CO1	3
Question No - 2	CO1	4
Question No - 3	CO3	6



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SET5

1. Change your host computer's ip to 10.0.0.5
2. You are trying to access www.google.com, but it's not loading; to check if the domain is resolving to the correct IP address, which command would help you verify the DNS resolution?
3. Write a TCP socket-based client-server C program where an IBM intern sends "Need Onboarding Help" and the server responds "IBM Helpdesk Activated" using port 8088, buffer size 600 bytes, and server IP 10.0.0.5.

Question Number.	Course Outcome (CO)	Taxonomy levels
Question No-1	CO1	3
Question No - 2	CO6	3
Question No - 3	CO3	6



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SET6

1. Change your host computer's ip to 172.16.5.10
2. Two computers on your local LAN can't communicate, and you want to confirm if the MAC address for the IP 192.168.0.12 is correctly cached in your system; which command will show the IP-to-MAC mappings?
3. Create a TCP client-server C program where a TCS employee sends "Start Secure Session" and the company server replies "TCS Secure Session Started" using port 9191, buffer size 800 bytes, and server IP 172.16.5.10.

Question Number.	Course Outcome (CO)	Taxonomy levels
Question No-1	CO1	3
Question No - 2	CO1	4
Question No - 3	CO3	6